

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12405539
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Baltis; Coen Hubertus Matheus et al.

Substrate support and substrate table

Abstract

A substrate support for supporting a substrate in a lithographic apparatus, the substrate support including: a support body configured to support the substrate; a main body separate from the support body and configured to support the support body, the main body including a thermal conditioner configured to thermally condition the main body and/or support body and/or substrate; and an extractor body surrounding the main body and the support body, the extractor body having an extraction channel configured to extract fluid from near a peripheral part of the substrate.

Inventors:	Baltis; Coen Hubertus Matheus (Eindhoven, NL), Ten Kate; Nicolaas (Almkerk, NL), Vermeulen; Marcus Martinus Petrus Adrianus (Leende, NL), Tromp; Siegfried Alexander (Knegsel, NL), Van Den Berkmortel; Frank Pieter Albert (Deurne, NL), Roset; Niek Jacobus Johannes (Eindhoven, NL), Kramer; Gijs (Nijmegen, NL), Brantjes; Nicolaas Petrus Marcus (Eindhoven, NL), Fonteyn; Michiel Theodorus Jacobus (Waalre, NL)
Applicant:	ASML NETHERLANDS B.V. (Veldhoven, NL)
Family ID:	74194761
Assignee:	ASML Netherlands B.V. (Veldhoven, NL)
Appl. No.:	17/798354
Filed (or PCT Filed):	January 25, 2021
PCT No.:	PCT/EP2021/051581
PCT Pub. No.:	WO2021/170320
PCT Pub. Date:	September 02, 2021

Prior Publication Data

Foreign Application Priority Data

EP	20158919	Feb. 24, 2020
EP	20163571	Mar. 17, 2020

Publication Classification**Int. Cl.:** G03F7/00 (20060101); H01L21/687 (20060101)**U.S. Cl.:****CPC** G03F7/707 (20130101); G03F7/70341 (20130101); G03F7/70808 (20130101); G03F7/70875 (20130101); H01L21/6875 (20130101); H01L21/68785 (20130101)**Field of Classification Search****USPC:** None

References Cited**U.S. PATENT DOCUMENTS**

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6952253	12/2004	Lof et al.	N/A	N/A
9354528	12/2015	Bex et al.	N/A	N/A
10168621	12/2018	Dekkers et al.	N/A	N/A
10509335	12/2018	Nakajima et al.	N/A	N/A
2002/0074635	12/2001	Hattori et al.	N/A	N/A
2004/0207824	12/2003	Lof et al.	N/A	N/A
2006/0102277	12/2005	Zaal et al.	N/A	N/A
2007/0109521	12/2006	Nishii	355/72	G03F 7/70341
2007/0146664	12/2006	Zaal et al.	N/A	N/A
2007/0153244	12/2006	Zaal et al.	N/A	N/A
2007/0242242	12/2006	Nagasaka	355/53	G03F 7/707
2007/0258080	12/2006	Brink et al.	N/A	N/A
2008/0158526	12/2007	Hennus et al.	N/A	N/A
2009/0296068	12/2008	Castelijns	355/72	G03F 7/70716
2010/0045949	12/2009	Nakano et al.	N/A	N/A
2011/0181849	12/2010	Patel	29/592	G03F 7/70875
2011/0222033	12/2010	Ten Kate et al.	N/A	N/A
2011/0228248	12/2010	Lafarre et al.	N/A	N/A
2012/0212725	12/2011	Lafarre et al.	N/A	N/A
2012/0274920	12/2011	Bex et al.	N/A	N/A
2012/0307216	12/2011	Laurent et al.	N/A	N/A
2013/0189802	12/2012	Tromp et al.	N/A	N/A
2015/0077728	12/2014	Laurent et al.	N/A	N/A
2015/0109599	12/2014	Koevoets	355/73	G03F 7/707

2015/0124234	12/2014	Lafarre et al.	N/A	N/A
2017/0312826	12/2016	Lafarre et al.	N/A	N/A
2018/0107107	12/2017	Kramer	N/A	G03F 7/70716
2018/0231897	12/2017	Jeunink	N/A	G03F 7/707
2018/0267412	12/2017	Laurent et al.	N/A	N/A
2019/0163066	12/2018	Kramer	N/A	G03F 7/707
2019/0235395	12/2018	Lafarre et al.	N/A	N/A
2020/0142325	12/2019	Ten Kate et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2261743	12/2011	EP	N/A
1420298	12/2012	EP	N/A
2006-148101	12/2005	JP	N/A
2008-251754	12/2007	JP	N/A
2009-043879	12/2008	JP	N/A
2010-206113	12/2009	JP	N/A
2011-151397	12/2010	JP	N/A
2011-205093	12/2010	JP	N/A
2015-519755	12/2014	JP	N/A
201633000	12/2015	TW	N/A
99/49504	12/1998	WO	N/A
2013/178438	12/2012	WO	N/A
2013/178484	12/2012	WO	N/A
2015/064613	12/2016	WO	N/A
2017/102162	12/2016	WO	N/A

OTHER PUBLICATIONS

International Search Report dated Mar. 26, 2021, issued in corresponding International Application No. PCT/EP2021/051581 (3 pgs.). cited by applicant

Written Opinion of the International Searching Authority dated Mar. 26, 2021, issued in corresponding International Application No. PCT/EP2021/051581 (5 pgs.). cited by applicant

Primary Examiner: Thomas; Binu

Attorney, Agent or Firm: Pillsbury Winthrop Shaw Pittman, LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is the U.S. national phase entry of PCT Patent Application No. PCT/EP2021/051581 filed on Jan. 25, 2021, which claims the benefit of priority of European Patent Application No. 20158919.9 filed on Feb. 24, 2020 and of European Patent Application No. 20163571.1 filed on Mar. 17, 2020 which are incorporated herein in their entireties by reference.

FIELD

(2) The present invention relates to a substrate support for supporting a substrate in a lithographic apparatus, a substrate table, a method for supporting a substrate and a device manufacturing method.

BACKGROUND

(3) A lithographic apparatus is a machine constructed to apply a desired pattern onto a substrate. A lithographic apparatus can be used, for example, in the manufacture of integrated circuits (ICs). A lithographic apparatus may, for example, project a pattern (also often referred to as “design layout” or “design”) of a patterning device (e.g., a mask) onto a layer of radiation-sensitive material (resist) provided on a substrate (e.g., a wafer). Known lithographic apparatus include so-called steppers, in which each target portion is irradiated by exposing an entire pattern onto the target portion at one time, and so-called scanners, in which each target portion is irradiated by scanning the pattern through a radiation beam in a given direction (the “scanning”-direction) while synchronously scanning the substrate parallel or anti-parallel to this direction.

(4) As semiconductor manufacturing processes continue to advance, the dimensions of circuit elements have continually been reduced while the amount of functional elements, such as transistors, per device has been steadily increasing over decades, following a trend commonly referred to as ‘Moore's law’. To keep up with Moore's law the semiconductor industry is chasing technologies that enable to create increasingly smaller features. To project a pattern on a substrate a lithographic apparatus may use electromagnetic radiation. The wavelength of this radiation determines the minimum size of features which are patterned on the substrate. Typical wavelengths currently in use are 365 nm (i-line), 248 nm, 193 nm and 13.5 nm.

(5) Further improvements in the resolution of smaller features may be achieved by providing an immersion fluid having a relatively high refractive index, such as water, on the substrate during exposure. The effect of the immersion fluid is to enable imaging of smaller features since the exposure radiation will have a shorter wavelength in the fluid than in gas. The effect of the immersion fluid may also be regarded as increasing the effective numerical aperture (NA) of the system and also increasing the depth of focus.

(6) The immersion fluid may be confined to a localized area between the projection system of the lithographic apparatus and the substrate by a fluid handling structure. Fast relative movement between the substrate and the confined immersion liquid may cause leaking of the immersion fluid from the localized area. Such leaking is undesirable and may lead to defects on the substrate. The speed at which the substrate is stepped or scanned with respect to the projection system is thus limited. This limits the throughput of the lithographic apparatus.

SUMMARY

(7) During a semiconductor manufacturing process, the substrate is supported on a substrate support. Over time, the substrate support wears out and needs to be replaced. It is an object of the present invention to provide a substrate support that is cheaper to maintain and that reduces the length of downtime required to service it when it is worn.

(8) According to the present invention, there is provided substrate support for supporting a substrate in a lithographic apparatus, the substrate support comprising:

(9) a first support body configured to support the substrate;

(10) a main body separate from the first support body and configured to support the first support body, the main body comprising a thermal conditioner configured to thermally condition the main body, and/or the support body and/or the substrate; and

(11) an extractor body surrounding the main body and the support body, wherein the extractor body comprises a first extraction channel configured to extract fluid from near a peripheral part of the substrate.

(12) According to the present invention, there is also provided a substrate table comprising: a substrate stage; and the substrate support.

(13) According to the present invention, there is also provided a method for supporting a substrate in a lithographic apparatus on a substrate support comprising:

(14) supporting the substrate on a first support body of the substrate support;

(15) supporting the first support body on a main body separate from the first support body;

(16) thermally conditioning the main body, and/or the support body and/or the substrate with a

thermal conditioner of the main body; and

(17) extracting fluid from near a peripheral part of the substrate through a first extraction channel of an extractor body surrounding the main body and the support body.

(18) According to the present invention, there is also provided a method for replacing a first support body of a substrate support for supporting a substrate in a lithographic apparatus, the method comprising:

(19) connecting a carrier plate to the first support body such that the carrier plate covers pin holes that extend through the first support body in a thickness direction of the first support body;

(20) extending a plurality of pins through respective pin holes such that the pins support the carrier plate on their ends, wherein the pins extend through a main body of the substrate support below the first support body; and

(21) controlling movement of the pins so as to lower the first support body onto the main body or raise the first support body away from the main body.

(22) According to the present invention, there is also provided a device manufacturing method using a lithographic apparatus, the method comprising:

(23) projecting a beam patterned by a patterning device onto a substrate while supporting the substrate with a substrate support; and

(24) performing the method for supporting the substrate in the lithographic apparatus or the method for replacing a first support body of a substrate support for supporting a substrate in a lithographic apparatus.

(25) Further embodiments, features and advantages of the present invention, as well as the structure and operation of the various embodiments, features and advantages of the present invention, are described in detail below with reference to the accompanying drawings.

Description

DESCRIPTION OF THE DRAWINGS

(1) Embodiments of the invention will now be described, by way of example only, with reference to the accompanying schematic drawings, in which corresponding reference symbols indicate corresponding parts, and in which:

(2) FIG. 1 depicts a schematic overview of the lithographic apparatus;

(3) FIGS. 2 and 3 depict, in cross section, two different versions of a fluid handling system for use in a lithographic projection apparatus;

(4) FIG. 4 depicts part of a lithographic apparatus;

(5) FIG. 5 depicts a substrate support according to an embodiment of the invention;

(6) FIG. 6 depicts a substrate support according to a comparative example;

(7) FIGS. 7 to 11 depict different versions of a substrate support according to the invention;

(8) FIGS. 12 and 13 depict different versions of a height adjustment mechanism for an embodiment of the invention;

(9) FIG. 14 is a schematic cross-sectional view of a carrier plate connected to a first support body according to an embodiment of the invention;

(10) FIG. 15 is a schematic cross-sectional view of the assembly shown in FIG. 14 lowered onto a main body according to an embodiment of the invention;

(11) FIG. 16 is a schematic cross-sectional view of the carrier plate being separated from the first support body shown in FIGS. 14 and 15;

(12) FIG. 17 is a schematic cross-sectional view of a carrier plate connected to a first support body according to an embodiment of the invention;

(13) FIG. 18 is a schematic cross-sectional view of a carrier plate connected to a first support body according to an embodiment of the invention;

- (14) FIG. 19 is a schematic cross-sectional view of a carrier plate connected to a first support body according to an embodiment of the invention;
- (15) FIG. 20 is a schematic cross-sectional view of a mechanism for connecting a carrier plate to a first support body according to an embodiment of the invention;
- (16) FIG. 21 is a schematic cross-sectional view part way through a process of disconnecting the carrier plate from the first support body shown in FIG. 20;
- (17) FIG. 22 is a schematic cross-sectional view of the carrier plate disconnected from the first support body shown in FIG. 20;
- (18) FIG. 23 is a schematic cross-sectional view of a mechanism for connecting a carrier plate to a first support body according to an embodiment of the invention;
- (19) FIG. 24 is a schematic cross-sectional view part way through a process of disconnecting the carrier plate from the first support body shown in FIG. 23;
- (20) FIG. 25 is a schematic cross-sectional view of the carrier plate disconnected from the first support body shown in FIG. 23;
- (21) FIG. 26 is a schematic plan view of a first support body according to an embodiment of the invention;
- (22) FIG. 27 is a schematic plan view of a first support body according to an embodiment of the invention being gripped for handling; and
- (23) FIG. 28 is a schematic plan view of a first support body according to an embodiment of the invention being gripped for handling.
- (24) The features shown in the Figures are not necessarily to scale, and the size and/or arrangement depicted is not limiting. It will be understood that the Figures include optional features which may not be essential to the invention. Furthermore, not all of the features of the apparatus are depicted in each of the figures, and the Figures may only show some of the components relevant for describing a particular feature.

DETAILED DESCRIPTION

- (25) In the present document, the terms “radiation” and “beam” are used to encompass all types of electromagnetic radiation, including ultraviolet radiation (e.g. with a wavelength of 365, 248, 193, 157 or 126 nm).
- (26) The term “reticle”, “mask” or “patterning device” as employed in this text may be broadly interpreted as referring to a generic patterning device that can be used to endow an incoming radiation beam with a patterned cross-section, corresponding to a pattern that is to be created in a target portion of the substrate. The term “light valve” can also be used in this context. Besides the classic mask (transmissive or reflective, binary, phase-shifting, hybrid, etc.), examples of other such patterning devices include a programmable mirror array and a programmable LCD array.
- (27) FIG. 1 schematically depicts a lithographic apparatus. The lithographic apparatus includes an illumination system (also referred to as illuminator) IL configured to condition a radiation beam B (e.g., UV radiation or DUV radiation), a mask support (e.g., a mask table) MT constructed to support a patterning device (e.g., a mask) MA and connected to a first positioner PM configured to accurately position the patterning device MA in accordance with certain parameters, a substrate support (e.g., a substrate table) WT constructed to hold a substrate (e.g., a resist coated wafer) W and connected to a second positioner PW configured to accurately position the substrate support WT in accordance with certain parameters, and a projection system (e.g., a refractive projection lens system) PS configured to project a pattern imparted to the radiation beam B by patterning device MA onto a target portion C (e.g., comprising one or more dies) of the substrate W.
- (28) In operation, the illumination system IL receives the radiation beam B from a radiation source SO, e.g. via a beam delivery system BD. The illumination system IL may include various types of optical components, such as refractive, reflective, magnetic, electromagnetic, electrostatic, and/or other types of optical components, or any combination thereof, for directing, shaping, and/or controlling radiation. The illuminator IL may be used to condition the radiation beam B to have a

desired spatial and angular intensity distribution in its cross section at a plane of the patterning device MA.

(29) The term “projection system” PS used herein should be broadly interpreted as encompassing various types of projection system, including refractive, reflective, catadioptric, anamorphic, magnetic, electromagnetic and/or electrostatic optical systems, or any combination thereof, as appropriate for the exposure radiation being used, and/or for other factors such as the use of an immersion liquid or the use of a vacuum. Any use of the term “projection lens” herein may be considered as synonymous with the more general term “projection system” PS.

(30) The lithographic apparatus is of a type wherein at least a portion of the substrate W may be covered by an immersion liquid having a relatively high refractive index, e.g., water, so as to fill an immersion space 11 between the projection system PS and the substrate W—which is also referred to as immersion lithography. More information on immersion techniques is given in U.S. Pat. No. 6,952,253, which is incorporated herein by reference.

(31) The lithographic apparatus may be of a type having two or more substrate supports WT (also named “dual stage”). In such “multiple stage” machine, the substrate supports WT may be used in parallel, and/or steps in preparation of a subsequent exposure of the substrate W may be carried out on the substrate W located on one of the substrate support WT while another substrate W on the other substrate support WT is being used for exposing a pattern on the other substrate W.

(32) In addition to the substrate support WT, the lithographic apparatus may comprise a measurement stage (not depicted in figures). The measurement stage is arranged to hold a sensor and/or a cleaning device. The sensor may be arranged to measure a property of the projection system PS or a property of the radiation beam B. The measurement stage may hold multiple sensors. The cleaning device may be arranged to clean part of the lithographic apparatus, for example a part of the projection system PS or a part of a system that provides the immersion liquid. The measurement stage may move beneath the projection system PS when the substrate support WT is away from the projection system PS.

(33) In operation, the radiation beam B is incident on the patterning device, e.g. mask, MA which is held on the mask support MT, and is patterned by the pattern (design layout) present on patterning device MA. Having traversed the mask MA, the radiation beam B passes through the projection system PS, which focuses the beam onto a target portion C of the substrate W. With the aid of the second positioner PW and a position measurement system IF, the substrate support WT can be moved accurately, e.g., so as to position different target portions C in the path of the radiation beam B at a focused and aligned position. Similarly, the first positioner PM and possibly another position sensor (which is not explicitly depicted in FIG. 1) may be used to accurately position the patterning device MA with respect to the path of the radiation beam B. Patterning device MA and substrate W may be aligned using mask alignment marks M.sub.1, M.sub.2 and substrate alignment marks P.sub.1, P.sub.2. Although the substrate alignment marks P.sub.1, P.sub.2 as illustrated occupy dedicated target portions, they may be located in spaces between target portions. Substrate alignment marks P.sub.1, P.sub.2 are known as scribe-lane alignment marks when these are located between the target portions C.

(34) To clarify the invention, a Cartesian coordinate system is used. The Cartesian coordinate system has three axis, i.e., an x-axis, a y-axis and a z-axis. Each of the three axis is orthogonal to the other two axis. A rotation around the x-axis is referred to as an Rx-rotation. A rotation around the y-axis is referred to as an Ry-rotation. A rotation around about the z-axis is referred to as an Rz-rotation. The x-axis and the y-axis define a horizontal plane, whereas the z-axis is in a vertical direction. The Cartesian coordinate system is not limiting the invention and is used for clarification only. Instead, another coordinate system, such as a cylindrical coordinate system, may be used to clarify the invention. The orientation of the Cartesian coordinate system may be different, for example, such that the z-axis has a component along the horizontal plane.

(35) Immersion techniques have been introduced into lithographic systems to enable improved

resolution of smaller features. In an immersion lithographic apparatus, a liquid layer of immersion liquid having a relatively high refractive index is interposed in the immersion space **11** between a projection system PS of the apparatus (through which the patterned beam is projected towards the substrate W) and the substrate W. The immersion liquid covers at least the part of the substrate W under a final element of the projection system PS. Thus, at least the portion of the substrate W undergoing exposure is immersed in the immersion liquid.

(36) In commercial immersion lithography, the immersion liquid is water. Typically the water is distilled water of high purity, such as Ultra-Pure Water (UPW) which is commonly used in semiconductor fabrication plants. In an immersion system, the UPW is often purified and it may undergo additional treatment steps before supply to the immersion space **11** as immersion liquid. Other liquids with a high refractive index can be used besides water as the immersion liquid, for example: a hydrocarbon, such as a fluorohydrocarbon; and/or an aqueous solution. Further, other fluids besides liquid have been envisaged for use in immersion lithography.

(37) In this specification, reference will be made in the description to localized immersion in which the immersion liquid is confined, in use, to the immersion space **11** between the final element and a surface facing the final element. The facing surface is a surface of substrate W or a surface of the supporting stage (or substrate support WT) that is co-planar with the surface of the substrate W. (Please note that reference in the following text to surface of the substrate W also refers in addition or in the alternative to the surface of the substrate support WT, unless expressly stated otherwise; and vice versa). A fluid handling structure IH present between the projection system PS and the substrate support WT is used to confine the immersion liquid to the immersion space **11**. The immersion space **11** filled by the immersion liquid is smaller in plan than the top surface of the substrate W and the immersion space **11** remains substantially stationary relative to the projection system PS while the substrate W and substrate support WT move underneath.

(38) Other immersion systems have been envisaged such as an unconfined immersion system (a so-called 'All Wet' immersion system) and a bath immersion system. In an unconfined immersion system, the immersion liquid covers more than the surface under the final element. The liquid outside the immersion space **11** is present as a thin liquid film. The liquid may cover the whole surface of the substrate W or even the substrate W and the substrate support WT co-planar with the substrate W. In a bath type system, the substrate W is fully immersed in a bath of immersion liquid.

(39) The fluid handling structure IH is a structure which supplies the immersion liquid to the immersion space **11**, removes the immersion liquid from the immersion space **11** and thereby confines the immersion liquid to the immersion space **11**. It includes features which are a part of a fluid supply system. The arrangement disclosed in PCT patent application publication no. WO 99/49504 is an early fluid handling structure comprising pipes which either supply or recover the immersion liquid from the immersion space **11** and which operate depending on the relative motion of the stage beneath the projection system PS. In more recent designs, the fluid handling structure extends along at least a part of a boundary of the immersion space **11** between the final element of the projection system PS and the substrate support WT or substrate W, so as to in part define the immersion space **11**.

(40) The fluid handling structure IH may have a selection of different functions. Each function may be derived from a corresponding feature that enables the fluid handling structure IH to achieve that function. The fluid handling structure IH may be referred to by a number of different terms, each referring to a function, such as barrier member, seal member, fluid supply system, fluid removal system, liquid confinement structure, etc.

(41) As a barrier member, the fluid handling structure IH is a barrier to the flow of the immersion liquid from the immersion space **11**. As a liquid confinement structure, the structure confines the immersion liquid to the immersion space **11**. As a seal member, sealing features of the fluid handling structure IH form a seal to confine the immersion liquid to the immersion space **11**. The sealing features may include an additional gas flow from an opening in the surface of the seal

member, such as a gas knife.

(42) In an embodiment the fluid handling structure IH may supply immersion fluid and therefore be a fluid supply system.

(43) In an embodiment the fluid handling structure IH may at least partly confine immersion fluid and thereby be a fluid confinement system.

(44) In an embodiment the fluid handling structure IH may provide a barrier to immersion fluid and thereby be a barrier member, such as a fluid confinement structure.

(45) In an embodiment the fluid handling structure IH may create or use a flow of gas, for example to help in controlling the flow and/or the position of the immersion fluid.

(46) The flow of gas may form a seal to confine the immersion fluid so the fluid handling structure IH may be referred to as a seal member; such a seal member may be a fluid confinement structure.

(47) In an embodiment, immersion liquid is used as the immersion fluid. In that case the fluid handling structure IH may be a liquid handling system. In reference to the aforementioned description, reference in this paragraph to a feature defined with respect to fluid may be understood to include a feature defined with respect to liquid.

(48) A lithographic apparatus has a projection system PS. During exposure of a substrate W, the projection system PS projects a beam of patterned radiation onto the substrate W. To reach the substrate W, the path of the radiation beam B passes from the projection system PS through the immersion liquid confined by the fluid handling structure IH between the projection system PS and the substrate W. The projection system PS has a lens element, the last in the path of the beam, which is in contact with the immersion liquid. This lens element which is in contact with the immersion liquid may be referred to as ‘the last lens element’ or “the final element”. The final element is at least partly surrounded by the fluid handling structure IH. The fluid handling structure IH may confine the immersion liquid under the final element and above the facing surface.

(49) As depicted in FIG. 1, in an embodiment the lithographic apparatus comprises a controller 500. The controller 500 is configured to control the substrate table WT.

(50) FIG. 2 schematically depicts a localized liquid supply system or fluid handling system. The liquid supply system is provided with a fluid handling structure IH (or liquid confinement structure), which extends along at least a part of a boundary of the space 11 between the final element of the projection system PS and the support table WT or substrate W. The fluid handling structure IH is substantially stationary relative to the projection system PS in the XY plane though there may be some relative movement in the Z direction (in the direction of the optical axis). In an example, a seal is formed between the fluid handling structure IH and the surface of the substrate W and may be a contactless seal such as a gas seal (such a system with a gas seal is disclosed in EP1,420,298) or liquid seal.

(51) The fluid handling structure IH at least partly confines the immersion liquid in the space 11 between the final element of the projection system PS and the substrate W. The space 11 is at least partly formed by the fluid handling structure IH positioned below and surrounding the final element of the projection system PS. Immersion liquid is brought into the space 11 below the projection system PS and within the fluid handling structure IH by one of liquid openings 13. The immersion liquid may be removed by another of liquid openings 13. The immersion liquid may be brought into the space 11 through at least two liquid openings 13. Which of liquid openings 13 is used to supply the immersion liquid and optionally which is used to remove the immersion liquid may depend on the direction of motion of the support table WT.

(52) The immersion liquid may be confined in the space 11 by a contactless seal such as a gas seal 16 formed by a gas which, during use, is formed between the bottom of the fluid handling structure IH and the surface of the substrate W. The gas in the gas seal 16 is provided under pressure via inlet 15 to the gap between the fluid handling structure IH and substrate W. The gas is extracted via outlet 14. The overpressure on the gas inlet 15, vacuum level on the outlet 14 and geometry of the gap are arranged so that there is a high-velocity gas flow inwardly that confines the immersion

liquid. Such a system is disclosed in US 2004/0207824, which is hereby incorporated by reference in its entirety. In an example, the fluid handling structure IH does not have the gas seal 16.

(53) FIG. 3 is a side cross sectional view that depicts a further liquid supply system or fluid handling system according to an embodiment. The arrangement illustrated in FIG. 3 and described below may be applied to the lithographic apparatus described above and illustrated in FIG. 1. The liquid supply system is provided with a fluid handling structure IH (or a liquid confinement structure), which extends along at least a part of a boundary of the space 11 between the final element of the projection system PS and the support table WT or substrate W.

(54) The fluid handling structure IH at least partly confines the immersion liquid in the space 11 between the final element of the projection system PS and the substrate W. The space 11 is at least partly formed by the fluid handling structure IH positioned below and surrounding the final element of the projection system PS. In an example, the fluid handling structure IH comprises a main body member 53 and a porous member 33. The porous member 33 is plate shaped and has a plurality of holes (i.e., openings or pores). In an embodiment, the porous member 33 is a mesh plate wherein numerous small holes 84 are formed in a mesh. Such a system is disclosed in US 2010/0045949 A1, which is hereby incorporated by reference in its entirety.

(55) The main body member 53 comprises supply ports 72, which are capable of supplying the immersion liquid to the space 11, and a recovery port 73, which is capable of recovering the immersion liquid from the space 11. The supply ports 72 are connected to a liquid supply apparatus 75 via passageways 74. The liquid supply apparatus 75 is capable of supplying the immersion liquid to the supply ports 72 through the corresponding passageway 74. The recovery port 73 is capable of recovering the immersion liquid from the space 11. The recovery port 73 is connected to a liquid recovery apparatus 80 via a passageway 79. The liquid recovery apparatus 80 recovers the immersion liquid recovered via the recovery port 73 through the passageway 29. The porous member 33 is disposed in the recovery port 73. Performing the liquid supply operation using the supply ports 72 and the liquid recovery operation using the porous member 33 forms the space 11 between the projection system PS and the fluid handling structure IH on one side and the substrate W on the other side.

(56) FIG. 4 illustrates part of a lithographic apparatus useful for an understanding of an embodiment of the present invention. The arrangement illustrated in FIG. 4 and described below may be applied to the lithographic apparatus described above and illustrated in FIG. 1. FIG. 4 is a cross-section through a substrate support 20 and a substrate W. In an embodiment, the substrate support 20 comprises one or more conditioning channels 61 of a thermal conditioner 60, which is described in more detail below. A gap 5 exists between an edge of the substrate W and an edge of the substrate support 20. When the edge of the substrate W is being imaged or at other times such as when the substrate W first moves under the projection system PS (as described above), the immersion space 11 filled with liquid by the fluid handling structure IH (for example) will pass at least partly over the gap 5 between the edge of the substrate W and the edge of the substrate support 20. This can result in liquid from the immersion space 11 entering the gap 5.

(57) The substrate W is held by a first support body 21 (e.g. a pimple or burl table) comprising one or more projections 41 (i.e., burls). The first support body 21 is an example of an object holder. Another example of an object holder is a mask holder. An under-pressure applied between the substrate W and the substrate support 20 helps ensure that the substrate W is held firmly in place. However, if immersion liquid gets between the substrate W and the first support body 21 this can lead to difficulties, particularly when unloading the substrate W.

(58) In order to deal with the immersion liquid entering that gap 5 at least one drain 10, 12 is provided at the edge of the substrate W to remove immersion liquid which enters the gap 5. In the embodiment of FIG. 4 two drains 10, 12 are illustrated though there may only be one drain or there could be more than two drains. In an embodiment, each of the drains 10, 12 is annular so that the whole periphery of the substrate W is surrounded.

(59) A primary function of the first drain **10** (which is radially outward of the edge of the substrate W/first support body **21**) is to help prevent bubbles of gas from entering the immersion space **11** where the liquid of the fluid handling structure IH is present. Such bubbles may deleteriously affect the imaging of the substrate W. The first drain **10** is present to help avoid gas in the gap **5** escaping into the immersion space **11** in the fluid handling structure IH. If gas does escape into the immersion space **11**, this can lead to a bubble which floats within the immersion space **11**. Such a bubble, if in the path of the projection beam, may lead to an imaging error. The first drain **10** is configured to remove gas from the gap **5** between the edge of the substrate W and the edge of the recess in the substrate support **20** in which the substrate W is placed. The edge of the recess in the substrate support **20** may be defined by a cover ring **101** which is optionally separate from the first support body **21** of the substrate support **20**. The cover ring **101** may be shaped, in plan, as a ring and surrounds the outer edge of the substrate W. The first drain **10** extracts mostly gas and only a small amount of immersion liquid.

(60) The second drain **12** (which is radially inward of the edge of the substrate W/first support body **21**) is provided to help prevent liquid which finds its way from the gap **5** to underneath the substrate W from preventing efficient release of the substrate W from the substrate table WT after imaging. The provision of the second drain **12** reduces or eliminates any problems which may occur due to liquid finding its way underneath the substrate W.

(61) As depicted in FIG. 4, in an embodiment the lithographic apparatus comprises a first extraction channel **102** for the passage therethrough of a two phase flow. The first extraction channel **102** is formed within a block. The first and second drains **10**, **12** are each provided with a respective opening **107**, **117** and a respective extraction channel **102**, **113**. The extraction channel **102**, **113** is in fluid communication with the respective opening **107**, **117** through a respective passageway **103**, **114**.

(62) As depicted in FIG. 4, the cover ring **101** has an upper surface. The upper surface extends circumferentially around the substrate W on the first support body **21**. In use of the lithographic apparatus, the fluid handling structure IH moves relative to the substrate support **20**. During this relative movement, the fluid handling structure IH moves across the gap **5** between the cover ring **101** and the substrate W. In an embodiment the relative movement is caused by the substrate support **20** moving under the fluid handling structure IH. In an alternative embodiment the relative movement is caused by the fluid handling structure IH moving over the substrate support **20**. In a further alternative embodiment the relative movement is provided by movement of both the substrate support **20** under the fluid handling structure IH and movement of the fluid handling structure IH over the substrate support **20**. In the following description, movements of the fluid handling structure IH will be used to mean the relative movement of the fluid handling structure IH relative to the substrate support **20**.

(63) FIG. 5 is a schematic diagram of a substrate support **20** according to an embodiment of the invention. The substrate support **20** is for supporting a substrate W in the lithographic apparatus (such as the lithographic apparatus shown in FIG. 1). During an exposure process, the substrate W is supported on the substrate support **20**. Between exposure processes, the substrate W on the substrate support **20** may be replaced. During this time, the substrate support **20** may not have any substrate W supported on it.

(64) As shown in FIG. 5, in an embodiment the substrate support **20** comprises a first support body **21**. The first support body **21** is configured to support the substrate W. When the substrate W is supported by the substrate support **20**, the substrate W comes into direct contact with the first support body **21**. The first support body **21** is the part of the substrate support **20** that physically supports the underside of the substrate W. As shown in FIG. 5, in an embodiment the first support body **21** comprises a plurality of burls **41**. The distal ends of the burls **41** form a plane at which the underside of the substrate W is supported. The underside of the substrate W comes into contact with the distal ends of the burls **41**. The burls **41** are at the upper side of the first support body **21**.

(65) As shown in FIG. 5, in an embodiment there is a space **42** between the underside of the substrate **W** and the base surface (between the burls **41**) at the upper side of the first support body **21**. In an embodiment, the substrate support **20** comprises a clamp configured to hold the substrate **W** onto the first support body **21**. For example, as shown in FIG. 5, in an embodiment the substrate support **20** comprises a clamp comprising a clamp channel **43**. The clamp channel **43** is configured to extract gas from the space **42** between the substrate **W** and the first support body **21**. The clamp channel **43** is in fluid communication with the space **42**. As shown in FIG. 5, in an embodiment the clamp channel **43** is in fluid communication with the space **42** via one of more clamp passageways **44**. In an embodiment, the clamp channel **43** has an annular shape around the center of the substrate support **20**. The clamp channel **43** extends circumferentially. In an embodiment, a plurality of clamp passageways **44** extend vertically to connect the clamp channel **43** to the space **42**.

(66) As shown in FIG. 5, in an embodiment the substrate support **20** comprises a main body **22**. The main body **22** is separate from the first support body **21**. The first support body **21** can be separated from the main body **22** without substantially damaging the main body **22**. The first support body **21** can be removed from, and placed back onto, the main body **22**. The main body **22** is configured to support the first support body **21**.

(67) During use of the substrate support **20**, the substrate support **20** undergoes wear and tear. For example, the distal ends of the burls **41** wear over time. When the burls **41** are worn, the first support body **21** can be replaced without having to replace the entirety of the substrate support **20**. For example, the first support body **21** may be replaced without replacing the main body **22**. In an embodiment, the first support body **21** is removed from the main body **22**. A replacement first support body **21** is then positioned on the main body **22**. Exposure processes are then continued using the replacement first support body **21**. An embodiment of the invention is expected to reduce the cost of maintaining usability of the substrate support **20**.

(68) It may be quicker to replace the first support body **21** compared to replacing the entirety of the substrate support **20**. An embodiment of the invention is expected to reduce the downtime of the lithographic apparatus required in order to maintain the substrate support **20**.

(69) As shown in FIG. 5, in an embodiment, the main body **22** comprises a thermal conditioner **60**. In an embodiment the thermal conditioner **60** is configured to thermally condition the main body **22**. Additionally, or alternatively, the thermal conditioner **60** is configured to thermally condition the first support body **21**. Additionally or alternatively, the thermal conditioner **60** is configured to thermally condition the substrate **W**. By thermally conditioning the main body **22** and/or the first support body **21** and/or the substrate **W**, the temperature profile can be controlled. In particular, deformations caused by thermal fluctuations can be reduced. By reducing deformations, the accuracy of the exposure processes can be improved.

(70) The way in which the thermal conditioner **60** thermally conditions is not particularly limited. Merely as an example, as shown in FIG. 5 in an embodiment the thermal conditioner **60** comprises one or more conditioning channels **61**. The conditioning channels **61** may extend through the main body **22**. In an embodiment, the conditioning channels **61** contain a fluid (e.g. a gas and/or a liquid).

(71) As shown in FIG. 1, in an embodiment, the lithographic apparatus comprises a controller **500**. In an embodiment the controller **500** is configured to control a flow or fluid through the one of more conditioning channels **61**. In an embodiment, the controller **500** is configured to control the flow of fluid through the conditioning channels **61** so as to control the temperature profile of a target body (e.g. the main body **22**, the first support body **21** and/or the substrate **W**).

(72) As shown in FIG. 5, in an embodiment the thermal conditioner **60** comprises one or more sensors **62**. In an embodiment, the sensor **62** is attached to the main body **22** or embedded in the main body **22**. In an embodiment, the sensor **62** is a temperature sensor. In an embodiment, the sensor **62** may be configured to sense the temperature of fluid flowing in the conditioning channels **61**. In an embodiment, a plurality of sensors **62** are provided at different positions along the

conditioning channels **61** so as to detect the temperature profile throughout the conditioning channels **61**.

(73) As shown in FIG. 5, in an embodiment the thermal conditioner **60** comprises one or more heaters **63**. As shown in FIG. 5, in an embodiment the heaters **63** are attached at the underside of the main body **22**. However, heaters **63** may be positioned at other surfaces of the main body **22** or the first support body **21**. Additionally or alternatively, sensors may be provided at the surfaces of the main body **22** and/or the first support body **21**. In an embodiment, the controller **500** is configured to receive signals from the sensors **62** and subsequently control the heaters **63** and/or a flow of fluid through the conditioning channels **61**.

(74) In the substrate support **20**, the function of physical supporting (and clamping) the substrate W is performed by the first support body **21**. Meanwhile, the function of thermally conditioning is performed by the main body **22**. The first support body **21** is separate from the main body **22**. The functions of supporting the substrate W and thermal stabilization are separated into different components. When one function fails (or is no longer performed to the required standard), then the corresponding component can be replaced without having to replace the other component. An embodiment of the invention is expected to make it easier to maintain the substrate support **20** so that the functions performed by it are performed to the required standard.

(75) As shown in FIG. 5, in an embodiment the substrate support **20** comprises an extractor body **100**. The extractor body **100** surrounds the main body **22** and the first support body **21**. The extractor body **100** is radially outwards of the main body **22** and the first support body **21**. As shown in FIG. 5, in an embodiment the extractor body **100** is provided in a component which is separate from the main body **22**. However, it is not essential for the extractor body **100** to be separate from the main body **22** and the first support body **21**. As will be described in more detail below, the extractor body **100** may be part of the same component as the main body **22**. In such an embodiment, the extractor body **100** cannot be replaced without also replacing the main body **22** (and vice versa). In an embodiment, for example, as shown in FIGS. 7 to 10, the extractor body **100** is part of the same component as the main body **22**. The extractor body **100** may be integrally formed with the main body **22**. The extractor body **100** and the main body **22** are formed together as one component. The extractor body **100** and the main body **22** cannot be disconnected from each other. In an alternative embodiment, the extractor body **100** is integrally formed with the substrate stage **26**.

(76) As shown in FIG. 5, in an embodiment the extractor body **100** comprises a first extraction channel **102**. In an embodiment, an extractor body **100** comprises the first drain **10** described above. The first extraction channel **102** is configured to extract fluid from near a peripheral part of the substrate W. As shown in FIG. 5, in an embodiment the first extraction channel **102** is configured to extract fluid from radially outwards of the peripheral edge of the substrate W. The first extraction channel **102** is configured to remove bubbles that may otherwise be present in the immersion liquid used in the lithographic apparatus. The first extraction channel **102** helps to reduce image defects by reducing bubbles and/or by reducing the generation of watermarks for example.

(77) As explained above, the extractor body **100** is configured to reduce defectivity issues. The function performed by the extractor body **100** may be different from the function performed by the first support body **21** and the main body **22**. When one of the functions fails, or is no longer performed to the required standard, the corresponding component (e.g. the extractor body **100** or the first support body **21** or the main body **22**) can be replaced without replacing the other components of the substrate support **20**.

(78) FIG. 6 schematically depicts a substrate support **20** according to a comparative example. In the comparative example shown in FIG. 6, both the function of supporting (and clamping) the substrate W and thermally conditioning the substrate W are performed in the same component. Accordingly, when the tips of the burls **41** wear out, the entire main body **22** including both of the thermal

conditioner **60** and the burls **41** that support the substrate **W** has to be replaced. This is necessary even if, for example, the thermal conditioner **60** is still in good order and does not need to be replaced.

(79) As shown in FIG. 5, in an embodiment a seal member **27** is provided to seal a gap between the substrate support **20** and the substrate stage **26** that supports the substrate support **20**. In the comparative example shown in FIG. 6, when the substrate support **20** is replaced, it is also necessary to remove and replace the seal member **27**. In contrast, in the embodiment shown in FIG. 5, the extractor body **100** and the seal member **27** may remain in place while the first support body **21** and/or main body **22** is replaced. Similarly, the embodiments in which the extractor body **100** is part of the same component as the main body **22**, the extractor body **100** and the seal member **27** can remain in place while the first support body **21** is replaced. An embodiment of the invention is expected to reduce the time required to maintain the substrate support **20**.

(80) As shown in FIG. 5, in an embodiment the substrate support **20** comprises a second support body **23**. The second support body **23** is separate from the main body **22**. The second support body **23** is configured to support the main body **22** on the substrate stage **26**. In an embodiment, the main body **22** comes into physical contact with the second support body **23**. For example, in an embodiment the underside of the main body **22** is supported on top of the second support body **23**. The second support body **23** is supported on the substrate stage **26**.

(81) As shown in FIG. 5, in an embodiment the second support body **23** comprises a plurality of burls **91**. In an embodiment, the burls **91** of the second support body **23** are longer than the burls **41** of the first support body **21**. The burls **91** protrude at the underside of the second support body **23**. These burls **91** extend towards the substrate stage **26** and are supported by the substrate stage **26**. In an embodiment, the second support body **23** is configured to reduce relative movement between the substrate support **20** and the substrate stage **26**. The second support body **23** is configured to reduce slipping between the substrate support **20** and the substrate stage **26**. During use of the lithographic apparatus, the substrate stage **20** may undergo high accelerations. If the substrate support **20** slips relative to the substrate stage **26**, then this can lead to undesirable overlay errors. The burls **91** are configured to grip relative to the substrate stage **26** so as to reduce slipping of the substrate support **20** relative to the substrate stage **26**.

(82) The second support body **23** can be physically separated from the main body **22** without damaging the main body **22** or the second support body **23**. It is possible to replace one or the other of the main body **22** and the second support body **23** without replacing the other. For example, if the anti-slip function of the second support body **23** is no longer being performed to the required standard, then the second support body **23** can be replaced without replacing the main body **22**. An embodiment of the invention is expected to reduce the cost for maintaining the substrate support **20**.

(83) In another example, the main body **22** may be replaced if, for example, the thermal conditioner **60** needs to be replaced. The main body **22** can be replaced without replacing (or even substantially moving) the second support body **23**. An embodiment of the invention is expected to reduce the cost of maintaining the substrate support **20**.

(84) However, it is not essential for the substrate support **20** to comprise such a second support body **23**. For example, as shown in FIGS. 7 to 12, the function of reducing slipping of the substrate support **20** relative to the substrate stage **26** may be performed by the main body **22**. The burls **91** may be provided at the bottom of the main body **22**.

(85) As shown in FIGS. 5 and 7, for example, in an embodiment the main body **22** comprises at its upper side a plurality of burls **65**. The burls **65** have distal ends in a plane. The burls **65** are configured to support the first support body **21**. By providing the burls **65** of the upper side of the main body **22**, the structure of the first support body **21** is kept simple. The underside of the first support body **21** can be flat and substantially featureless. The upper side of the first support body **21** comprises the burls **41** for supporting the substrate **W**. An embodiment of the invention is expected

to make it cheaper to manufacture the component that clamps the substrate W.

(86) Additionally, as shown in FIGS. 5 and 8 to 10, for example, in an embodiment the first support body **21** comprises a plurality of seal protrusions **46**. The seal protrusions **46** protrude at the upper side of the first support body **21**. The seal protrusions **46** are shorter than the burls **41**. The seal protrusions **46** do not come into contact with the substrate W. A small gap is present between the top of the seal protrusions **46** and the underside of the substrate W. In an embodiment, the seal protrusions **46** extend circumferentially around the first support body **21**, i.e., the seal protrusions **46** form respective rings. The seal protrusions **46** are configured to reduce the amount of fluid passing across the seal protrusions **46**.

(87) However, it is not essential for the main body **22** to be provided with the burls **65**. For example, in an alternative embodiment shown in FIG. 8, the first support body **21** comprises at its lowest side a plurality of burls **47**. The burls **47** have distal ends in a plane. The burls **47** are configured to contact the upper side of the main body **22**. As shown in FIG. 8, instead of the burls **65** being provided on the main body **22**, a different set of burls **47** may be provided at the bottom side of the first support body **21**. During use, it may be that the burls **41**, **47** wear out before other parts of the substrate support **20** wear out. By providing both sets of burls **41**, **47** on the first support body **21**, the burls can be replaced simply by replacing the first support body **21** without needing to replace the main body **22**. An embodiment of the invention is expected to reduce the cost of maintaining the substrate support **20**.

(88) As shown in FIG. 5, in an embodiment the extractor body **100** comprises a second extraction channel **113**. This is also shown in other Figures. The second extraction channel **113** is part of the second drain **12**. In an embodiment, the second extraction channel **113** is configured to extract fluid from radially inward of the first extraction channel **102**. In an embodiment, the second extraction channel **113** is provided to keep liquid from flowing under the substrate W. However, it is not essential for such a second extraction channel **113** to be provided. For example, FIG. 10 shows an embodiment that does not have the second extraction channel **113**.

(89) In an embodiment, the second extraction channel **113** is configured to extract fluid from below the peripheral part of the substrate W so as to prevent liquid from reaching the space **42** between the central part of the substrate W and the first support body **21**. As shown in FIG. 4, in an embodiment the second extraction channel **113** is connected to the space below the peripheral part of the substrate W via one or more second passageways **114**. In an embodiment, the second extraction channel **113** extends circumferentially around the substrate support **20**. In an embodiment, a plurality of second passageways **114** are provided. The second passageways **114** extend vertically between the second extraction channel **113** and just below the peripheral part of the substrate W.

(90) As shown in FIG. 5, in an embodiment the second extraction channel **113** is spaced apart from the first extraction channel **102** by an open gap. The first extraction channel **102** and the second extraction channel **113** may be provided in separate bodies **24**, **25**. The separate bodies **24**, **25** can be separated from each other without destroying either of the bodies **24**, **25**. The functions of reducing defectivity issues (performed by the first extraction channel **102**) and keeping liquid from flowing under the substrate W (performed by the second extraction channel **113**) may be performed by separate bodies **24**, **25**. It is possible to replace one of the bodies **24**, **25** without replacing the other. An embodiment of the invention is expected to reduce the cost of maintaining the substrate support **20**.

(91) As shown in FIG. 7 (and also visible in FIGS. 5 and 8, for example), in an embodiment the substrate support **20** comprises an inner seal **112** and an outer seal **111**. The inner seal **112** is positioned radially inward of the openings through which the second extraction channel **113** extracts fluid. The outer seal **111** is positioned radially outward of the openings through which the second extraction channel **113** extracts fluid. The inner seal **112** and the outer seal **111** protrude towards the underside of the substrate W. The inner seal **112** and the outer seal **111** do not protrude

far enough to come into contact with the substrate W in use with the substrate support **20**. Instead, a small gap is present between the tops of the inner seal **112** and the substrate W as well as between the top of the outer seal **111** and the substrate W. The inner seal **112** and the outer seal **11** are configured such that in use liquid may be present between the seal **111**, **112** and the substrate W. This helps to prevent liquid from reaching below the substrate W in a central part of the substrate W.

(92) As shown in FIG. **9**, in an embodiment the function of the inner seal **112** is performed by a seal protrusion **46** at the upper side of the first support body **21**. This is an alternative to the arrangement shown in FIG. **7**, for example, where the inner seal **112** is provided as a protrusion at the upper side of the main body **22**. As shown in FIG. **9**, in an embodiment an upper passageway **49** is provided through the first support body **21**. The upper passageway **49** connects with the second passage way **114** so that fluid can be extracted from below the peripheral substrate W into the second extraction channel **113**.

(93) As shown in FIG. **9**, for example, in an embodiment the main body **22** comprises a seal protrusions **66**. The seal protrusions **66** are configured to reduce the possibility of moisture reaching the upper surface of the main body **22**. This reduces oxidation of the main body **22**.

(94) As shown in FIG. **10**, in an embodiment the substrate support **20** is not provided with the second extraction channel **113** shown in FIGS. **5** to **9** for example. As shown in FIG. **10**, in an embodiment an open gap or a flow passage **115** is formed between the extractor body **100** and main body **22** and the first support body **21**. In an embodiment, the open gap or the flow passage **115** is in fluid communication with ambient pressure or a pressure source. In an embodiment, a plurality of flow passages **115** extend mainly vertically through the main body **22** (which may be part of the same component as the extractor body **100**). In an alternative embodiment, an open gap may extend circumferentially around the main body **22**, with the extractor body **100** provided as a separate component from the main body **22**. In an embodiment, a flow of gas passes through the open gap or flow passage **115**. The flow of gas passes over the seal protrusion **46** of the upper peripheral edge of the first support body **21**. By providing the gas flow, the possibility of liquid reaching the underside of the substrate W over the seal protrusion **46** may be reduced. It may not be necessary to provide the second extraction channel **113**. An embodiment of the invention is expected to reduce the complexity of the substrate support **20**. An embodiment of the invention is expected to make it easier to manufacture the substrate support **20**.

(95) As described above and shown in FIGS. **5** and **7** to **10**, the substrate support **20** can be modularized in various ways. As shown in FIG. **11**, in an embodiment an extractor body **100** is provided with both the first extraction channel **102** and the second extraction channel **113**. The extractor body **100** is separate from both the first support body **21** and the main body **22**. The main body **22** is supported directly on the substrate stage **26** (i.e. without the second support body **23**). As shown in FIG. **11**, the inner seal **112** and the outer seal **111** are provided as part of the extractor body **100**. This reduces the radial dimension of the first support body **21** (because the first support body **21** does not require an additional seal protrusion **46** to perform the function of the inner seal **112**). This reduces the amount of the substrate support **20** that is replaced when it is necessary to replace worn out burls **41**.

(96) As shown in FIG. **11**, in an embodiment the main body **22** extends below the extractor body **100**. The lower side of the extractor body **100** is coupled to the main body **22**. This makes it easier to control the height of the extractor body **100** relative to the first support body **21**. By controlling the height of the extractor body **100** more precisely, the sealing function provided by the inner seal **112** and the outer seal **111** is more reliable.

(97) As shown in FIG. **11**, in an embodiment one or more holes **119** are provided in the main body **22**. The holes **119** are configured to be in fluid communication with the gap between the radially inner part of the extractor body **100** and the main body **22**. The holes **119** may connect to other holes extending through the substrate stage **26**. The holes **119** are configured to provide ambient or

pressurized gas. By providing a gas flow up through the holes **119**, the possibility of liquid reaching below the central part of the substrate **W** is reduced. By providing the gas flow through the holes **119**, humidity transfer to the first support body **21** is reduced. By reducing humidity transfer to the first support body **21**, oxidation of the first support body **21** is reduced.

(98) In an embodiment, the holes **119** are configured to provide pressurized gas. By providing a pressurized flow of gas, it is possible to control lifting of the edge of the substrate **W**. By controlling lifting of the outer edge of substrate **W**, wear on the outermost burl **41** of the first support body **21** can be reduced. An embodiment of the invention is expected to reduce wear of the substrate support **20**.

(99) As shown in FIG. **11**, in an embodiment an adhesive layer **104** is provided between the extractor body **100** and the main body **22**. The adhesive layer **104** fixes the extractor body **100** to the main body **22**. This makes it easier to control the gap from the substrate **W** to the inner seal **112** and the outer seal **111** of the extractor body **100**.

(100) In an embodiment, beads are dispersed in the adhesive layer **104**. For example, glass beads may be dispersed in a glue. The beads can be chosen in size so as to provide the required height step between the tops of the inner/outer seals **111**, **112** and the top of the seal protrusion **66** at the radially outward part of the top surface of the main body **22**. In an embodiment, the beads have a diameter of, for example, 45 microns, 50 microns or 55 microns.

(101) As shown in FIG. **11**, in an embodiment fluid connection is maintained from the first/second extraction channels **102**, **113** through the main body **22** (on which the extractor body **100** is attached) and subsequently through the substrate stage **26**. The fluid connection may be maintained by dog bone connectors **105** and rings **106**. However, the connection between the extractor body **100** and the main body **22** for the purposes of the first/second extraction channels **102**, **113** is not particularly limited.

(102) It is not essential for the main body **22** to extend below the extractor body **100**. In an alternative embodiment, the lower side of the extractor body **100** is attached to the substrate stage **26**. In an embodiment, the substrate table **WT** (which comprises the substrate stage **26** and the substrate support **20**) comprises a height adjustment mechanism **130**. Alternative versions of a height adjustment mechanism **130** are shown in FIGS. **12** and **13**, for example. The height adjustment mechanism **130** is configured to provide for fine-tunable height adjustment so as to obtain the right seal gap between the extractor body **100** and the substrate **W**. The height adjustment mechanism **130** is configured to control the height of at least part of the extractor body **100** below the substrate **W** such that the extractor body **100** is configured to prevent liquid from reaching between the central part of the substrate **W** and the first support body **21**.

(103) As shown in FIG. **12**, in an embodiment the height adjustment mechanism **130** comprises an insert member **131**. The insert member **131** is a block of material. The insert member **131** is provided between the lower side for the extractor body **100** and the substrate stage **26**. The insert member **131** is separate from the main body **22**. The insert member **131** is the component through which the extractor body **100** is attached to the substrate stage **26**.

(104) As shown in FIG. **12**, in an embodiment the height adjustment mechanism **130** comprises a fastener **132**. For example, in an embodiment the fastener **132** is a bolt. As shown in FIG. **12**, in an embodiment the fastener **132** does not reach through to the substrate stage **26**. The fastener **132** connects the extractor body **100** to the insert member **131**.

(105) As shown in FIG. **12**, in an embodiment the extractor body **100** comprises protrusions **134** configured to engage with the insert member **131**. As shown in FIG. **12**, in an embodiment the insert member **131** is glued onto the substrate stage **26**.

(106) As shown in FIG. **12**, in an embodiment the insert member **131** is structured with cut-out sections **133** such that the insert member **131** compresses when the fastener **132** fastens the extractor body **100** to the insert member **131**. The compression of the insert member **131** can be controlled so as to control the height of the top of the extractor body **100** below the substrate **W**. In

particular, the force supplied by the fastener **132** can be controlled so as to control compression of the insert member **131**.

(107) FIG. **12** shows the force line **135** of reaction forces that pass through the fastener **132** and through solid parts (i.e. between cut-out sections **133**) of the insert member **131**. As shown in FIG. **12**, the cut-out sections **133** may be arranged so that the force line **135** extends through a relatively narrow section of the insert member **131**. The narrow section of the insert member **131** may be relatively flexible, thereby making it easier for the insert member **131** to be compressed in a controlled way. This helps to control the height of the extractor body **100** below the substrate **W**. In particular, narrow sections of the insert member **131** between cut-out sections **133** may function as leaf springs (or other flexures).

(108) The height adjustment mechanism **130** shown in FIG. **12** is only one example. As another example, an alternative height adjustment mechanism **130** is shown in FIG. **13**. As shown in FIG. **13**, in an embodiment the extractor body **100** comprises a leaf spring **141**. The leaf spring **141** comprises a lower part **142** fixed to the rest of the extractor body **100**. The leaf spring **141** further comprises an upper part **143** that extends between the extractor body **100** and the first support body **21**. In an embodiment, a plurality of such leaf springs **141** are provided circumferentially around the substrate support **20**. In an alternative embodiment, a single leaf spring **141** extends circumferentially around the substrate support **20**.

(109) As shown in FIG. **13**, in an embodiment the height adjustment mechanism **130** is configured to control the height of the upper part **143** below the substrate **W** such that the upper part **143** is configured to prevent liquid from reaching between a central part of the substrate **W** and the first support body **21**. In an embodiment, a set screw **144** is provided to allow for fine adjustment of the leaf spring **141**. The set screw **144** can protrude a controlled distance below the extractor body **100**. As the set screw **144** protrudes further from the bottom of the extractor body **100**, the set screw **144** forces the leaf spring **141** to bend, thereby lowering the height of the upper part **144** of the leaf spring **141**. As shown in FIG. **13**, in an embodiment the first extraction channel **102** may be connected with the second extraction channel **113** via a connecting channel **140**.

(110) The way in which the thermal conditioner **60** is arranged in the main body **22** is not particularly limited. In an embodiment, the thermal conditioner **60** is attached at an under surface of the main body **22**. For example, one of more Peltier elements and/or heaters may be provided at the lower surface of the main body **22** and/or the extractor body **100**. Additionally or alternatively, the channels **61**, one of more Peltier elements and/or heaters **63** may be provided externally around the bottom of the main body **22**. In an alternative embodiment the thermal conditioner **60** is attached at an upper side of the main body **22**. The external channels **61** and/or one of more Peltier elements and/or heaters **63** may be attached (e.g. glued onto the main body **22**). In an embodiment, the channels **61**, heaters **63** and/or sensor **62** are arranged in the substrate stage **26**. At least part of the thermal conditioning function is performed by the substrate stage **26**. This helps to simplify the design of the main body **22**. In an embodiment, the burls **91** are provided as part of the substrate stage **26**. The main body **22** is not required to comprise the burls **91**. The underside of the main body **22** may be substantially flat. In an embodiment, at least one sensor **62** is provided in a respective burl **91** of the substrate stage **26**. The sensor **62** is configured to sense a temperature of the main body **22**. The sensor **62** is positioned close to the main body **22**. In an embodiment, at least one heater **63** is provided at a surface of the substrate stage **26** facing the main body **22**. In an embodiment, the heater **63** is paired with a respective sensor **62**. The heater **63** is positioned adjacent to the respective sensor **62**. In an embodiment, the heater **63** is controlled based on output from the paired sensor **62**.

(111) As shown in FIG. **5**, in an embodiment the clamping channel **43** is in fluid communication with the space **42** via another space **45** between the main body **22** and the first support body **21**.

(112) As shown in FIG. **5**, in an embodiment the substrate support **20** is locked relative to the substrate stage **26** by at least one locking bolt **29**. In an embodiment, the locking bolt **29** is

configured to function as a safety lock that prevents the substrate support **20** from falling out from the substrate stage **26**. However, the burls **91** perform the function of preventing slipping of the substrate support **20** relative to the substrate stage **26**.

(113) As shown in FIG. 5, in an embodiment one of more pins **28** may be used to lower the substrate W onto the first support body **21**. The pins **28** extend through respective pin holes **121** in the substrate support **20** and support the substrate W so as to control the height of the substrate W above the first support body **21** during loading and unloading sequences.

(114) As shown in FIG. 7, in an embodiment the main body **22** is formed in two parts that are attached together at a bonding line **64**. For example, a bonding material may be used.

(115) As mentioned above, the first support body **21** can be replaced without also replacing the other parts of the substrate holder **20**. As shown in the Figures, the first support body **21** has a generally similar shape to that of the substrate W. In an embodiment, the lithographic apparatus comprises at least one handling tool configured to move the substrate W, for example to move the substrate W between different sections of the lithographic apparatus and/or to move the substrate W into or out from the lithographic apparatus. In an embodiment, the same handling tool(s) can be used to control movement of the first support body **21**.

(116) FIG. 14 is a schematic cross-sectional view of a carrier plate **120** coupled to a first support body **21** according to an embodiment of the invention. The carrier plate **120** is configured to be couplable to the first support body **21**. The carrier plate **120** is configured to be releasable (decouplable) from the first support body **21**. As shown in FIG. 14, in an embodiment the carrier plate **120** is configured to cover the pin holes **121** in the first support body **21** when the carrier plate **120** is coupled to the first support body **21**.

(117) As mentioned above, in an embodiment a plurality of pins **28** support the substrate W and can be controlled to control lowering of the substrate W onto the substrate support **20**. FIG. 15 shows the assembly of the carrier plate **120** and first support body **21** supported by the pins **28**. By providing that the carrier plate **120** covers the pin holes **121**, the pins **28** can effectively support the first support body **21** via the carrier plate **120** because the first support body **21** is effectively coupled to the carrier plate **120**. An embodiment of the invention is expected to make it easier and/or to replace the first support body **21**. An embodiment of the invention is expected to reduce downtime of the lithographic apparatus. It is not essential for the carrier plate **120** to be provided. In an alternative embodiment (described in more detail below), alternatives to the pins **28** are provided for controlling movement of the first support body **21**.

(118) The pins **28** can be controlled so as to lower the first support body **21** onto the rest of the substrate support **20** because the first support body **21** and the carrier plate **120** are coupled together. Once the first support body **21** is mounted onto the rest of the substrate support **20**, the carrier plate **120** can be removed. FIG. 16 schematically shows the carrier plate **120** decoupled and removed from the first support body **21**. In an embodiment, an overpressure is applied through opening **44** to help release the carrier plate **120** from the first support body **21**. The carrier plate **120** can be lifted away from the first support body **21** by the pins **28**. In an embodiment, the carrier plate **120** can then be moved using the same handling tool(s) used to handle the substrate W.

(119) As shown in FIGS. 14 to 16, in an embodiment the carrier plate **120** comprises a plurality of connection protrusions **122**. The connection protrusions **122** are configured to protrude from a base plate **125** of the carrier plate **120** towards the first support body **21**. The carrier plate **120** comprises the base plate **125** and the connection protrusions **122**. The base plate **125** is flat. In an embodiment the base plate **125** and the connection protrusions **122** are formed of the same material. The base plate **125** is formed integrally with the connection protrusions **122** to form the carrier plate **120**. As shown in FIG. 14, in an embodiment the connection protrusions **122** limit how close the base plate **125** can get to the first support body **21**. As shown in FIG. 14, in an embodiment the connection protrusions **122** have an abutment surface **123**. The abutment surface **123** is configured to abut the first support body **21** directly or via a bonding material.

(120) By providing the connection protrusions **122**, the distal ends of the burls **41** do not come into contact with the carrier plate **120**. The carrier plate **120** is supported by the first support body **21** between the burls **41** at the upper side of the first support body **21** with the distal ends of the burls **41** at the upper side of the first support body **21** being spaced from the carrier plate **120**. An embodiment of the invention is expected to prevent further wear of the burls **41**.

(121) FIG. **17** is a schematic view of a carrier plate **120** according to an embodiment of the invention. As depicted in FIG. **17**, in an embodiment the carrier plate **120** is sufficiently flat at its surface facing the first support body **21** at the distal ends of the burls **41** at the upper side of the first support body **21** supports the carrier plate **120** when the carrier plate **120** is coupled to the first support body **21**. As shown in FIG. **17**, in an embodiment the connection protrusions **122** do not limit how close the carrier plate **120** can get to the first support body **21**. Instead, the distal ends of the burls **41** support the carrier plate **120**. The distal ends of the burls **41** contact the base plate surface **124** either directly or via a bonding material.

(122) As shown in FIGS. **14** and **15**, in an embodiment only a part of the carrier plate **120** is brought into contact with the first support body **21**. This limits the adhesion force, thereby making separation of the carrier plate **120** and the first support body **21** easier.

(123) As shown in FIGS. **17** and **19**, for example, it is not essential for the carrier plate **120** to contact the surface of the first support body **21** between the burls **41**. In an embodiment, the surface between the burls **41** is rough, such that adhesion forces are limited. By not requiring contact with the rough surface between the burls **41**, the rough surface does not adversely affect the carrier plate **120** being coupled to the first support body **21**.

(124) In the arrangement shown in FIG. **17**, the distal ends of the burls **41** contact the carrier plate **120**. One reason for the first support body **21** being replaced is that the burls may have worn. The distal ends of the burls **41** may be particularly smooth when the first support body **21** is required to be replaced. The distal ends of the burls **41** may allow good adhesion to the carrier plate **120**, particularly for unloading a first support body **21**.

(125) The way that the carrier plate **120** is coupled to the first support body **21** is not particularly limited. FIG. **18** schematically shows use of a bonding material **126** to couple the carrier plate **120** to the first support body **21**. As shown in FIG. **18**, in an embodiment a bonding material **126** is provided between the first support body **21** and the abutment surface **123** of the connection protrusions **122**. Additionally or alternatively, bonding material **126** can be provided between the distal ends of the burls **41** and the base plate surface **124**.

(126) The bonding material **126** is for temporarily bonding the carrier plate **120** to the first support body **21** when the carrier plate **120** is coupled to the first support body **21**. The bonding material **126** is not particularly limited. In an embodiment, the bonding material **126** comprises an adhesive material. In an alternative embodiment, a layer of liquid (e.g. water) is provided in place of the bonding material **126**. Capillary pressure can be used to keep the carrier plate **120** coupled to the first support body **21**.

(127) In the arrangement shown in FIG. **17**, an adhesive material may be applied to the base plate surface **124** in the recesses. By limiting the adhesive material to the recesses, the possibility of the adhesive material undesirably contacting other components is reduced.

(128) As shown in FIG. **19**, additionally or alternatively, the carrier plate **120** is clamped to the first support body **21** by pressure. In an embodiment the carrier plate **120** comprises at least one internal channel **127**. The internal channel **127** is for fluidly connecting a first opening **128** to at least one second opening **129**. The first opening **128** and the second opening **129** are at the surface of the carrier plate **120**. As shown in FIG. **19**, in an embodiment the first opening **128** is at a position where the carrier plate **120** covers one of the pin holes **121** when the carrier plate **120** is coupled to the first support body **21**.

(129) The second opening **129** is at a position where the carrier plate **120** faces the first support body **21** at its upper side. In an embodiment, the first opening **128** can be connected to an

underpressure so as to cause a flow of gas through the internal channel **127** from the second opening **129** to the first opening **128** and out through the first opening **128**. This can reduce the pressure in the region between the carrier plate **120** and the first support body **21** when the carrier plate **120** is coupled to the first support body **21**. The carrier plate **120** and the first support body **21** can be held together as a result of the surrounding pressure being greater than the pressure between the carrier plate **120** and the first support body **21**.

(130) As shown in FIG. **19**, in an embodiment the first opening **128** is at a position that corresponds to the pin **28** that supports the carrier plate **120**. In an embodiment, the pin **28** comprises a channel for extracting fluid from the internal channel **127** through the first opening **128**. The number of second openings **129** is not particularly limited. The number of internal channels **127** is not particularly limited. In an embodiment, a first opening **128** is provided for each respective pin hole **121**.

(131) In an embodiment, the carrier plate **120** comprises an electrically conductive material and an insulating layer configured to electrically insulate the electrically conductive material from the first support body **21** when the carrier plate **120** is connected to the first support body **21**. The electrically conductive material can be at a potential difference relative to the first support body **21**. Electrostatic attraction between the carrier plate **120** and the first support body **21** can help to couple the carrier plate **120** to the first support body **21**. In an embodiment, the insulating layer is provided as a thin layer surrounding the carrier plate **120**.

(132) FIGS. **20** to **22** show different stages of a carrier plate **120** being released from a first support body **21** according to an embodiment of the invention. As shown in FIG. **20**, in an embodiment the carrier plate **120** comprises at least one connection mechanism **151**. The connection mechanism **151** is configured to mechanically lock the carrier plate **120** to the first support body **21**. In an embodiment, the connection mechanism **151** locks the carrier plate **120** to the first support body **21** at a respective pin hole **121** of the first support body **21**.

(133) As shown in FIG. **20**, in an embodiment the connection mechanism **151** comprises a chamber **153**. The chamber **153** is configured to receive a respective pin **28**. The chamber **153** receives the contacting end of the pin **28** when the pin **28** supports the carrier plate **120**. In an embodiment the connection mechanism **151** is configured to release the lock when pressure within the chamber **153** is sufficiently lower than ambient pressure outside of the chamber **153** such that the carrier plate **120** can be released from the first support body **21**. The carrier plate **120** can be released from the first support body **21** when the lock is released.

(134) As shown in FIG. **20**, in an embodiment the connection mechanism **151** comprises at least one locking element **152**. The locking element **152** is configured to extend through the pin hole **121** and engage with the surface of the first support body **21** that faces away from the base plate **125** of the carrier plate **120**. The number of locking elements **152** is not particularly limited. In an embodiment, two, three, four or more than four locking elements **152** are provided. The locking elements **152** may be evenly distributed circumferentially around the pin hole **121**. In an embodiment, the locking element **152** is substantially continuous circumferentially around the inner surface of the pin hole **121**. The locking element **152** is fixedly connected at one end to the base plate **125** of the carrier plate **120**. In an embodiment the locking element **152** is formed integrally with the base plate **125** of the carrier plate **120**.

(135) In an embodiment, the chamber **153** is connected to an underpressure so as to reduce the pressure in the chamber **153**. The chamber **153** is connected to the underpressure when the carrier plate **120** is to be released from the first support body **21**. As mentioned above, in an embodiment the pin **28** comprises a channel for extracting fluid from beyond the contacting end of the pin **28**. In an embodiment, the channel in the pin **28** is configured to extract fluid from the chamber **153** so as to reduce the pressure in the chamber **153**. As shown in FIG. **20**, in an embodiment a plurality of pores **154** are provided for fluidly connecting the channel of the pin **28** to the chamber **153**. The number and form of the pores **154** is not particularly limited. In an embodiment a mesh comprising

the pores **154** is provided. In an embodiment, the mesh is formed integrally with the base plate **125** of the carrier plate **120**. In an embodiment, the contacting end of the pin **28** is not completely perpendicular to the longitudinal direction of the pin **28** such that the channel within the pin **28** is fluidly connected with the chamber **153** when the pin **28** is supporting the carrier plate **120**.

(136) FIG. **21** shows the process step in which the pressure in the chamber **153** is reduced. As a result of the reduction in pressure in the chamber **153**, the locking elements **152** bend inwardly towards the pin **28**. This causes the locking elements **152** to cease engagement with the surface of the first support body **21** that faces away from the base plate **125** of the carrier plate **120**. When the locking elements **152** stop engaging with the first support body **21**, the carrier plate **120** is no longer locked to the first support body **21**. As shown in FIG. **22**, the carrier plate **120** can then be released from the first support body **21**.

(137) FIGS. **23** to **25** schematically show an alternative connection mechanism **151** according to an embodiment of the invention. As shown in FIG. **23**, in an embodiment the chamber **153** is defined by a housing **157**. In an embodiment, part of the housing **157** protrudes out from the surface of the carrier plate **120** that faces away from the first support body **21**. In an embodiment, the housing **157** is formed as a separate component from the base plate **125** and is subsequently connected to the base plate **125**. An embodiment of the invention is expected to make it easier to manufacture the carrier plate **120**. The design of the base plate **125** is made more simple.

(138) In an embodiment, the material used for the housing **157** is different from the material used for the base plate **125** of the carrier plate **120**. As shown in FIG. **23**, in an embodiment the locking element **152** is formed integrally with the housing **157** that defines the chamber **153**.

(139) As shown in the FIG. **23**, in an embodiment the carrier plate **120** comprises an unlock channel part **155**. The unlock channel part **155** defines an unlock channel through which fluid can flow for unlocking the carrier plate **120** from the first support body **21**. As shown in FIG. **23**, in an embodiment the carrier plate **120** comprises a resilient member **156** configured to connect the housing **157** to the unlock channel part **155**. In an embodiment the unlock channel part **155** is part of the base plate **125** of the carrier plate **120**.

(140) As shown in FIG. **24**, in an embodiment the channel within the pin **28** is connected to the unlock channel within the unlock channel part **155**. This allows the pressure in the chamber **153** to be reduced. When the pressure within the chamber **153** is reduced, the housing **157** is forced downwards relative the base plate **125** of the carrier plate **120**. As shown in FIG. **24**, in an embodiment the housing **157** comprises a section angled relative to a surface of the base plate **125** such that downwards movement of the housing **157** relative to the base plate **125** causes the locking element **152** to bend. When the locking element **152** bends, the locking element **152** ceases to engage with the first support body **21**. When the locking element **152** no longer engages with the first support body **21**, the carrier plate **120** is unlocked from the first support body **21**. The resilient member **156** can be compressed when the pressure in the chamber **153** is reduced. Otherwise, the resilient member **156** maintains a higher position of the housing **157** relative the base plate **125** of the carrier plate **120** as shown in FIG. **23**. As shown in FIG. **25**, when the carrier plate **120** is unlocked, the carrier plate **120** can be released from the first support body **21**.

(141) It is not essential for a carrier plate **120** to be provided. FIG. **26** is a schematic plan view of a first support body **21** according to an embodiment of the invention. As shown in FIG. **26**, in an embodiment no carrier plate **120** is provided. As a result, the pins **28** can extend freely through the respective pin holes **121**. In an embodiment the pins **28** are not used to lower the first support body **21** onto the main body **22** of the substrate support **20**. In an embodiment, an additional set of pins **160** are provided at different locations from the pins **28** that are used to lower the substrate **W**. The outlines of the tops of the pins **160** are shown in FIG. **26**. In an embodiment, a further set of pin holes for accommodating the pins **160** are provided in the parts of the substrate support **20** lower than the first support body **21**. For example, such a further set of pin holes is provided in the main body **22**. The pins **160** extend through the further set of pin holes in the main body **22** and support

the lower surface of the first support body **21**. Meanwhile, the pins **28** for supporting the substrate **W** extend through the pin holes **121** that extend through both the main body **22** and the first support body **21**.

(142) In an alternative embodiment, the pins **28** that are used for lowering the substrate **W** support the first support body **21** when the first support body **21** has a rotational position different from its target position on the main body **22**. The first support body **21** can be supported by the pins **28** and lowered onto the main body **22** in its rotated orientation. The carrier plate **120** is not required because the pin holes **121** through the first support body **21** do not line up with the pins **28**. This is because the first support body **21** is rotated relative to its final target position. Once the first support body **21** is supported by the main body **22**, the first support body **21** is then rotated so as to reach its target rotational position on the main body **22**. When the first support body **21** is in its target position on the main body **22**, the pin holes **121** line up between the first support body **21** and the main body **22** such that the pins **28** can extend through the pin holes **121** so as to support the substrate **W** above the first support body **21**. In an embodiment, a rotation tool is provided to orient the first support body **21** rotationally on the main body **22**. By rotating the first support body **21** once it is supported by the main body **22**, it is not necessary to provide a carrier plate **120** and it is not necessary to provide a separate set of pins **160**.

(143) FIG. **27** is a schematic plan view of a first support body **21** according to an embodiment of the invention. In the view shown in FIG. **27**, the first support body **21** is being gripped for handling. As shown in the FIG. **27**, in an embodiment a plurality of edge grippers **161** are provided for gripping the first support body **21**. The edge grippers **161** are configured to grip the outer edge of the first support body **21**. In an embodiment, the edge grippers **161** are comprised in a separate handling tool for handling the first support body **21**. By providing a separate handling tool, it is not necessary to use the pins **28** to lower the first support body **21** and it is not necessary to provide a carrier plate **120**.

(144) FIG. **28** is a schematic plan view of a first support body **21** according to an embodiment of the invention. In the view shown in FIG. **28**, the first support body **21** is being gripped for handling. As shown in FIG. **28**, in an embodiment a top gripper **162** is provided for gripping the top of the first support body **21**. By providing the top gripper **162**, it is not necessary to provide a carrier plate **120** or to provide a separate set of pins **160** for handling the support body **21**. In an embodiment, the top gripper **162** is comprised in a separate handling tool for handling the first support body **21**. The separate handling tool is separate from the pins **28** that are used for lowering the substrate **W** onto the first support body **21**.

(145) Features shown in the different embodiments illustrated in the Figures can be combined with each other unless it is apparent that they are incompatible. Merely as an example, the feature of providing burls **47** at the underside of the first support body **21** (shown in FIG. **8**) may be applied to the arrangement shown in FIG. **5**, **9**, **10** or **11**. In particular, the arrangement shown in FIG. **9** could be modified by replacing the burls **65** which are at the upper side of the main body **22** with burls **47** being provided at the underside of the first support body **21**. Other combinations of features shown in the Figures are equally possible.

(146) Embodiments are provided according to the following clauses:

(147) 1. A substrate support for supporting a substrate in a lithographic apparatus, the substrate support comprising: a first support body configured to support the substrate; a main body separate from the first support body and configured to support the first support body, the main body comprising a thermal conditioner configured to thermally condition the main body, and/or the first support body and/or the substrate; and an extractor body surrounding the main body and the first support body, wherein the extractor body comprises a first extraction channel configured to extract fluid from near a peripheral part of the substrate.

(148) 2. The substrate support of clause 1, comprising: a second support body separate from the main body and configured to support the main body on a substrate stage, the second support body

configured to reduce relative movement between the substrate support and the substrate stage.

(149) 3. The substrate support of clause 1 or 2, wherein the main body comprises at its upper side a plurality of burls having distal ends in a plane configured to support the first support body or wherein the first support body comprises at its lower side a plurality of burls having distal ends in a plane configured to contact an upper side of the main body.

(150) 4. The substrate support of any of clauses 1 to 3, wherein the extractor body comprises a second extraction channel configured to extract fluid from radially inward of the first extraction channel, preferably wherein the second extraction channel is configured to extract fluid from below the peripheral part of the substrate so as to prevent liquid from reaching a space between a central part of the substrate and the first support body, and preferably wherein the second extraction channel is spaced apart from the first extraction channel by an open gap.

(151) 5. The substrate support of any of clauses 1 to 4, wherein an open gap or a flow passage is formed between the extractor body and the main body and the first support body, or wherein the extractor body forms an integral part of the main body.

(152) 6. The substrate support of clause 5, wherein the open gap or the flow passage is in fluid communication with ambient pressure or a pressure source, and/or wherein the main body extends below the extractor body and a lower side of the extractor body is coupled to the main body, preferably wherein an adhesive layer is provided between the extractor body and the main body.

(153) 7. The substrate support of any preceding clauses, wherein thermal conditioner is attached at an under surface of the main body, and/or wherein pin holes extend through the first support body in a thickness direction of the first support body, the pin holes configured such that respective pins can extend through the pin holes to lower a substrate onto the first support body.

(154) 8. The substrate support of clause 7, comprising a carrier plate configured to be couplable to and releasable from the first support body, the carrier plate being configured such that when the carrier plate is coupled to the first support body, the carrier plate covers the pin holes.

(155) 9. The substrate support of clause 8, wherein: the first support body comprises at its upper side a plurality of burls having distal ends in a plane configured to support the substrate; and the carrier plate comprises connection protrusions configured such that the carrier plate is supported by the first support body between the burls at the upper side of the first support body with the distal ends of the burls at the upper side of the first support body being spaced from the carrier plate, or wherein: the first support body comprises at its upper side a plurality of burls having distal ends in a plane configured to support the substrate; and the carrier plate is sufficiently flat at its surface facing the first support body that the distal ends of the burls at the upper side of the first support body support the carrier plate when the carrier plate is coupled to the first support body.

(156) 10. The substrate support of clause 8 or 9, comprising: a bonding material for temporarily bonding the carrier plate to the first support body when the carrier plate is coupled to the first support body, and/or wherein the carrier plate comprises at least one internal channel for fluidly connecting a first opening to at least one second opening, the first opening being at a position where the carrier plate covers one of the pin holes when the carrier plate is coupled to the first support body, and the second opening being at a position where the carrier plate faces the first support body at its upper side, and/or wherein the carrier plate comprises an electrically conductive material and an insulating layer configured to electrically insulate the electrically conductive material from the first support body when the carrier plate is coupled to the first support body, and/or wherein the carrier plate comprises at least one connection mechanism configured to mechanically lock the carrier plate to the first support body at a respective pin hole of the first support body.

(157) 11. The substrate support of clause 10, wherein the connection mechanism comprises a chamber configured to receive a respective pin, wherein the connection mechanism is configured to release the lock when pressure within the chamber is sufficiently lower than ambient pressure outside of the chamber such that the carrier plate can be released from the first support body.

- (158) 12. A substrate table comprising: a substrate stage; and the substrate support of clause 5 or 6, wherein a lower side of the extractor body is attached to the substrate stage.
- (159) 13. The substrate table of clause 12, comprising: a height adjustment mechanism configured to control a height of at least part of the extractor body below the substrate such that the extractor body is configured to prevent liquid from reaching between a central part of the substrate and the first support body.
- (160) 14. The substrate table of clause 13, wherein the height adjustment mechanism comprises: an insert member between the lower side of the extractor body and the substrate stage; and a fastener configured to fasten the extractor body to the insert member, wherein the insert member is structured with cut-out sections such that the insert member compresses when the fastener fastens the extractor body to the insert member so as to control the height of the extractor body below the substrate, or wherein: the extractor body comprises a leaf spring comprising: a lower part fixed to the rest of the extractor body; and an upper part that extends between the extractor body and the first support body; and the height adjustment mechanism is configured to control the height of the upper part below the substrate such that the upper part is configured to prevent liquid from reaching between a central part of the substrate and the first support body.
- (161) 15. A substrate table comprising: a substrate stage; and the substrate support of any of clauses 1 to 4, wherein the extractor body is provided in the substrate stage.

Claims

1. A substrate support for supporting a substrate in a lithographic apparatus, the substrate support comprising: a first support body configured to support the substrate; a main body separate from the first support body and configured to removably support the first support body, the main body comprising a thermal conditioner configured to thermally condition the main body and/or first support body and/or substrate; and an extractor body surrounding the main body and the first support body, wherein the extractor body comprises a first extraction channel configured to extract fluid from near a peripheral part of the substrate.
2. The substrate support of claim 1, comprising a second support body separate from the main body and configured to support the main body on a substrate stage, the second support body configured to reduce relative movement between the substrate support and the substrate stage.
3. The substrate support of claim 1, wherein the main body comprises at its upper side a plurality of burls having distal ends in a plane configured to support the first support body or wherein the first support body comprises at its lower side a plurality of burls having distal ends in a plane configured to contact an upper side of the main body.
4. The substrate support of claim 1, wherein the extractor body comprises a second extraction channel configured to extract fluid from inward, relative to a central part of the substrate support, of the first extraction channel.
5. The substrate support of claim 1, wherein an open gap or a flow passage is formed between the extractor body and the main body and between the extractor body and the first support body.
6. The substrate support of claim 5, wherein the open gap or the flow passage is in fluid communication with ambient pressure or a pressure source.
7. The substrate support of claim 1, wherein pin holes extend through the first support body in a thickness direction of the first support body, the pin holes configured to allow respective pins to extend through respective pin holes to lower a substrate onto the first support body.
8. The substrate support of claim 7, further comprising a carrier plate configured to be couplable to and releasable from the first support body, the carrier plate configured such that when the carrier plate is coupled to the first support body, the carrier plate covers the pin holes.
9. The substrate support of claim 8, wherein: the first support body comprises at its upper side a plurality of burls having distal ends in a plane configured to support the substrate; and the carrier

plate comprises connection protrusions configured such that the carrier plate is supported by the first support body between the plurality of burls at the upper side of the first support body with the distal ends of the plurality of burls at the upper side of the first support body being spaced from the carrier plate, or wherein: the first support body comprises at its upper side a plurality of burls having distal ends in a plane configured to support the substrate; and the carrier plate has a surface configured to face the first support body so that the distal ends of the plurality of burls at the upper side of the first support body support the carrier plate when the carrier plate is coupled to the first support body.

10. The substrate support of claim 8 wherein the carrier plate comprises at least one connection mechanism configured to mechanically lock the carrier plate to the first support body at a respective pin hole of the first support body.

11. The substrate support of claim 10, wherein the at least one connection mechanism comprises a chamber configured to receive a respective pin, wherein the connection mechanism is configured to release the lock when pressure within the chamber is lower than ambient pressure outside of the chamber to allow the carrier plate to be released from the first support body.

12. A substrate table comprising: a substrate stage; and the substrate support of claim 5, wherein a lower side of the extractor body is attached to the substrate stage.

13. The substrate table of claim 12, further comprising a height adjustment mechanism configured to control a height of at least part of the extractor body below the substrate such that the extractor body is configured to prevent liquid from reaching between a central part of the substrate and the first support body.

14. The substrate table of claim 13, wherein: the height adjustment mechanism comprises: an insert member between the lower side of the extractor body and the substrate stage; and a fastener configured to fasten the extractor body to the insert member, wherein the insert member is structured with cut-out sections such that the insert member compresses when the fastener fastens the extractor body to the insert member so as to control the height of the extractor body below the substrate, or the extractor body comprises a leaf spring comprising: a lower part fixed to the rest of the extractor body; and an upper part that extends between the extractor body and the first support body; and the height adjustment mechanism is configured to control the height of the upper part below the substrate such that the upper part is configured to prevent liquid from reaching between a central part of the substrate and the first support body.

15. A substrate table comprising: a substrate stage; and the substrate support of claim 1, wherein the extractor body is provided in the substrate stage.

16. The substrate support of claim 1, wherein the extractor body forms an integral part of the main body.

17. The substrate support of claim 1, wherein the main body extends below the extractor body and a lower side of the extractor body is coupled to the main body.

18. The substrate support of claim 1, wherein the thermal conditioner is attached at an under surface of the main body.

19. The substrate support of claim 8, wherein the carrier plate comprises an electrically conductive material and an insulating layer configured to electrically insulate the electrically conductive material from the first support body when the carrier plate is coupled to the first support body.

20. The substrate support of claim 8, wherein the carrier plate comprises at least one internal channel for fluidly connecting a first opening to at least one second opening, the first opening being at a position where the carrier plate covers one of the pin holes when the carrier plate is coupled to the first support body, and the second opening being at a position where the carrier plate faces the first support body at its upper side.
